

Mathematical Statistics 202: Assignment Term 4

Instructions

- Answer all the questions.
 - Please type up the solutions in a word processor of your choice (or learn to use RMarkdown).
 - Remember to include on your report your **name** (in a large font) and **e-mail** address on the front page. You could indicate your name **on every page** of your submitted document, i.e. **edit the header**.
 - You **MUST** label each answer by question number and where a question has multiple parts, **label each part of the question CLEARLY**.
 - On completion of your assignment, upload to RUConnected. Please submit as a word/open office file and a pdf. You are also required to submit your R code (i.e. the script file).
 - Acknowledge any help you may have received. Feel free to note any help you may have given to other students in the course.
 - This assignment must be submitted by Monday the 21th of September by 11.55 pm. Late submissions are to be emailed to Dr Chinomona and will be penalized.
 - All data in these assignments have been randomly generated. Please note that while it might be appropriate for you to use a t-test with equal variances for question 1 (say), it might not be the case for any or all of your class mates. In addition, you might make use of the same test statistic more than once in this assignment, your colleagues might not.
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1. A librarian is interested in the “profile” of students who make use of the Department of Physical Science library at Rhodes University. He randomly sampled students who use the library regularly. One of the variables in this study was the degree for which students were registered. The information he collected is shown in the following table: [10]

BA BSc BCom BCom BA BA BA BCom BA BCom BCom BA BA BSc

- (a) Capture these data into R, in a *suitable* variable called *degree*. Remember to submit this data file as part of your assignment. Please name the file **yourstudentnumber_question1.Rdata**
 - (b) Construct suitable numerical and graphical summaries for these data.
2. A Rhodes University student is interested in perceptions of “Rhodes” as a brand. He randomly samples students at Rhodes University. The non-Rhodes respondents were randomly selected people, who have not studied at Rhodes and who resided in his home town which is far away from Grahamstown. He asks each respondent the following question “Rhodes University is a brand that is distinct from the person Cecil John Rhodes”. Each respondent is asked to indicate if they either “Strongly disagree”, “Disagree”, “Are not certain”, “Agree” or “Strongly Agree”. The data can be found in the file *yourstudentnumber_question2_.csv*. [10]

- (a) Import the data into R.
 - (b) Create appropriate graphical and numerical summaries of these data.
 - (c) Do these data provide sufficient evidence that there is a relationship between a persons response to the question and if they are Rhodes Students or not Rhodes Students? Clearly justify your choice of test statistic and compose and appropriate conclusion.
3. Ward and Quinn (1988) investigated the differences in the fecundity, as measured by egg production (the **Fecundity** variable in the data set), of a predatory inter-tidal gastropod *Lepsiella vinosa* in two different inter-tidal zones, namely the mussel zone and the higher littorinid zone. A friend of yours in the Zoology II class has repeated the experiment as part of a tutorial/practical. The data can be found in the file yourstudentnumber_question3_.csv. [10]
- (a) Import the data into R.
 - (b) Create appropriate graphical and numerical summaries of these data.
 - (c) Do these data provide sufficient evidence that the egg production of the gastropod *Lepsiella vinosa* is significantly different in the different tidal zones. Clearly justify your choice of test statistic and compose an appropriate conclusion.
4. Furness and Bryant (1996) measured the metabolic rates, expressed in kilojoules (kJ), of male and female breeding northern fulmars. A friend of yours in the Zoology II class has repeated the experiment as part of a tutorial/practical. The data can be found in the file yourstudentnumber_question4_.csv. [10]
- (a) Import the data into R.
 - (b) Create appropriate graphical and numerical summaries of these data.
 - (c) Do these data provide sufficient evidence that there is a significant difference between the metabolic rates between the sexes? Clearly justify your choice of test statistic and compose and appropriate conclusion.
5. AZT was the first FDA-approved antiretroviral drug used in the care of HIV infected people (Source <http://www.aids.org>). The p24 antigen can stimulate an immune response and hence is monitored in patients who receive AZT. The measurements of p24 levels for patients who receive either the 300 mg or the 600 mg dosage are contained in the data file yourstudentnumber_question5_.csv. [10]
- (a) Create appropriate graphical and numerical summaries of these data.
 - (b) Do these data provide sufficient evidence that there is a significant difference between the average p24 levels between the two different dosages? Clearly justify your choice of test statistic and compose and appropriate conclusion.
6. Water-quality researchers at Rhodes University have measured the biomass to chlorophyll ratio for phytoplankton in milligrams per liter of water. Two laboratory tests exist to measure this ratio, one that is more expensive than the other. Each of the water samples are subjected to both tests. The results of these tests are in the data file yourstudentnumber_question6_.csv. [10]
- (a) Import the data into R. Please note that NA denotes a missing value, i.e. no test was conducted.
 - (b) Create appropriate graphical and numerical summaries of these data.

- (c) Do these data provide sufficient evidence that there is a significant difference between the average biomass to chlorophyll ratio for phytoplankton of these two laboratories? Clearly justify your choice of test statistic and compose and appropriate conclusion.
7. A postgraduate botany student is conducting an experiment on the fruit yields of a species of plant under various conditions, namely control, treatment 1 and treatment 2. You may assume that his experimental design is sound, that is that all the plants were identical at the start of this experiment etc. The data that this student collected can be found in the file `yourstudent-number_question7_.csv`. The yields are measured in kilograms (kg) and this data file has three columns of data, Control (No additional nutrients provided), Treatment1 (additional phosphates) and Treatment2 (additional nitrates). [10]
- (a) Import the data into R.
- (b) Create appropriate graphical and numerical summaries of these data.
- (c) Do these data provide sufficient evidence that there is a significant difference between the average yields for these three factors? Clearly justify your choice of test statistic and compose and appropriate conclusion. Hint: You may need to read up on a one-way analysis of variance.

END OF ASSIGNMENT

References

- Furness, R. W. and Bryant, D. M. (1996). Effect of wind on field metabolic rates of breeding northern fulmars. *Ecology*, 1181–1188.
- Ward, S. and Quinn, G. (1988). Preliminary investigations of the ecology of the intertidal predatory gastropod *Lepsiella vinosa* (Lamarck) (Gastropoda Muricidae). *Journal of Molluscan Studies*, 54(1), 109–117.